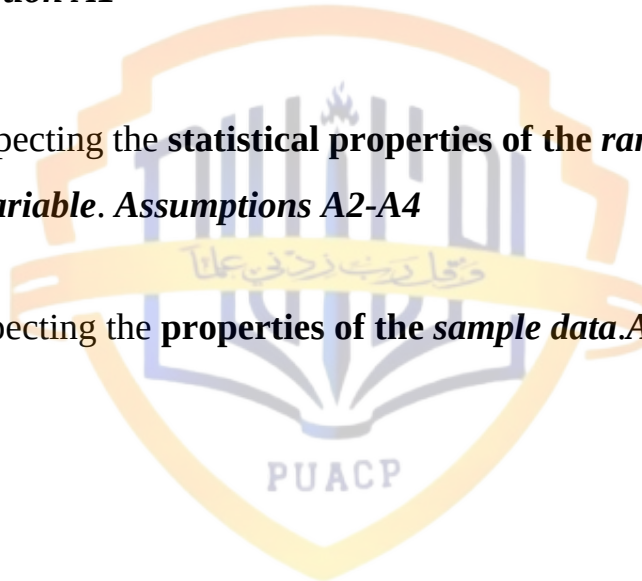

Specification -- Assumptions of the Simple Classical Linear Regression Model (CLRM)

1. Introduction

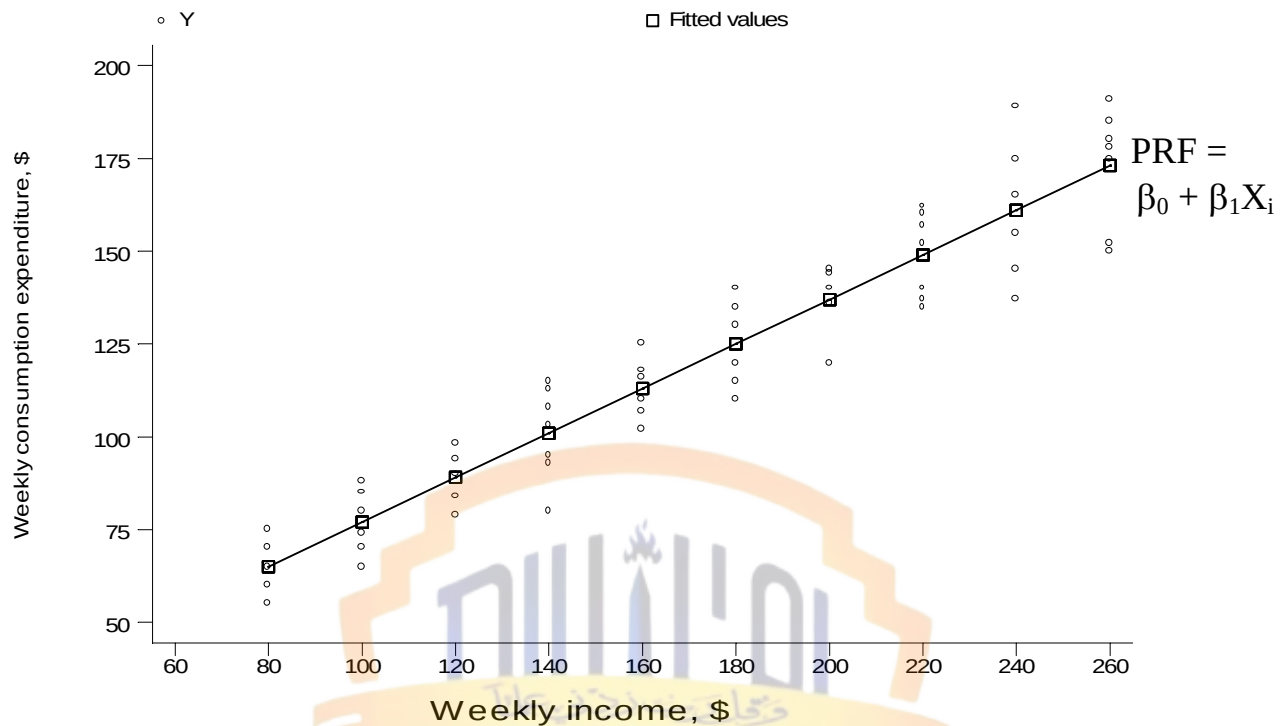
CLRM stands for the **Classical Linear Regression Model**. The CLRM is also known as the *standard* linear regression model.

Three sets of assumptions define the CLRM.

1. Assumptions respecting the **formulation of the *population regression equation*, or PRE. Assumption A1**
2. Assumptions respecting the **statistical properties of the *random error term* and the *dependent variable*. Assumptions A2-A4**
3. Assumptions respecting the **properties of the *sample data*. Assumptions A5-A8**



- **Figure 2.1** Plot of Population Data Points, Conditional Means $E(Y|X)$, and the Population Regression Function PRF



Recall that the solid line in Figure 2.1 is the population regression function, which takes the form $f(X_i) = E(Y_i | X_i) = \beta_0 + \beta_1 X_i$.

For each population value X_i of X , there is a **conditional distribution of population Y values** and a corresponding **conditional distribution of population random errors u_i** , where

(1) each population value of u for $X = X_i$ is

$$u_i | X_i = Y_i - E(Y_i | X_i) = Y_i - \beta_0 - \beta_1 X_i,$$

and

(2) each population value of Y for $X = X_i$ is

$$Y_i | X_i = E(Y_i | X_i) + u_i = \beta_0 + \beta_1 X_i + u_i.$$

2. Formulation of the Population Regression Equation (PRE)

Assumption A1: The population regression equation, or PRE, takes the form

$$Y = \beta_0 + \beta_1 X + u \quad \text{or} \quad Y_i = \beta_0 + \beta_1 X_i + u_i \quad (\text{A1})$$

- The PRE (A1) states that **each value** Y_i of the dependent variable Y can be written as **the sum of two parts**.
- 1. A linear function of the independent variable X that is called the **population regression function** (or **PRF**). The PRF for Y_i takes the form

$$f(X_i) = \beta_0 + \beta_1 X_i$$

- 2. A **random error term** u_i (also called a stochastic error term).

$$u_i = Y_i - f(X_i) = Y_i - (\beta_0 + \beta_1 X_i) = Y_i - \beta_0 - \beta_1 X_i$$

The random error terms are **unobservable** because the true population values of the regression coefficients β_0 and β_1 are unknown.

The PRE (A1) incorporates **three distinct assumptions**.

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad (\text{A1})$$

A1.1: Assumption of an Additive Random Error Term.

⇒ **The random error term u_i enters the PRE additively.**

Technically, this assumption means that the partial derivative of Y_i with respect to u_i equals 1: i.e.,

$$\frac{\partial Y_i}{\partial u_i} = 1 \quad \text{for all } i \text{ (} \forall i \text{)}.$$

A1.2: Assumption of Linearity-in-Parameters or Linearity-in-Coefficients.

⇒ **The PRE is linear in the population regression coefficients β_j ($j = 0, 1$).**

This assumption means that the partial derivative of Y_i with respect to each of the regression coefficients is a function only of known constants and/or the regressor X_i ; it is not a function of any unknown parameters.

$$\frac{\partial Y_i}{\partial \beta_j} = f_j(X_i) \quad j = 0, 1 \quad \text{where } f_j(X_i) \text{ contains no unknown parameters.}$$

A1.3: Assumption of Parameter or Coefficient Constancy.

⇒ **The population regression coefficients β_j ($j = 0, 1$) are (unknown) constants that do not vary across observations.**

$$\beta_{ji} = \beta_j = \text{a constant } \forall i \quad (j = 0, 1).$$

2.6. Determine whether the following models are linear in the parameters, or the variables, or both. Which of these models are linear regression models?

Model	Descriptive Title
a. $Y_i = \beta_1 + \beta_2 \left(\frac{1}{X_i} \right) + u_i$	Reciprocal
b. $Y_i = \beta_1 + \beta_2 \ln X_i + u_i$	Semilogarithmic
c. $\ln Y_i = \beta_1 + \beta_2 X_i + u_i$	Inverse semilogarithmic
d. $\ln Y_i = \ln \beta_1 + \beta_2 \ln X_i + u_i$	Logarithmic or double logarithmic
e. $\ln Y_i = \beta_1 - \beta_2 \left(\frac{1}{X_i} \right) + u_i$	Logarithmic reciprocal

Properties of the Random Error Term

Assumption A2: The Assumption of Zero Conditional Mean Error

The *conditional* mean, or *conditional* expectation, of the random error terms u_i for any given value X_i of the regressor X is equal to zero:

$$E(u|X) = 0 \quad \text{or} \quad E(u_i | X_i) = 0 \quad \forall i \quad (\text{for all } X_i \text{ values}) \quad (\text{A2})$$

This assumption says two things:

1. the **conditional** mean of the random error term u is the **same** for all **population values of X** -- i.e., it does not depend, either linearly or nonlinearly, on X ;
2. the common **conditional population mean of u** for all values of X is **zero**.

Implications of Assumption A2

- **Implication 1 of A2.** Assumption A2 implies that the **unconditional mean** of the population values of **the random error term u equals zero**:

$$E(u|X) = 0 \quad \Rightarrow \quad E(u) = 0 \quad (\text{A2-1})$$

or

$$E(u_i | X_i) = 0 \quad \Rightarrow \quad E(u_i) = 0 \quad \forall i. \quad (\text{A2-1})$$

This implication follows from the so-called **law of iterated expectations**, which states that $E[E(u|X)] = E(u)$. Since $E(u|X) = 0$ by A2, it follows that $E(u) = E[E(u|X)] = E[0] = 0$.

The logic of (A2-1) is straightforward: If the conditional mean of u for each and every population value of X equals zero, then the mean of these zero conditional means must also be zero.

- **Implication 2 of A2: the Orthogonality Condition.** Assumption A2 also implies that the population values X_i of the regressor X and u_i of the random error term u have *zero covariance* -- i.e., the population values of X and u are *uncorrelated*:

$$E(u|X) = 0 \Rightarrow \text{Cov}(X, u) = E(Xu) = 0 \quad (\text{A2-2})$$

or

$$E(u_i | X_i) = 0 \Rightarrow \text{Cov}(X_i, u_i) = E(X_i u_i) = 0 \quad \forall i \quad (\text{A2-2})$$

1. The equality $\text{Cov}(X_i, u_i) = E(X_i u_i)$ in (A2-2) follows from the definition of the covariance between X_i and u_i , and from assumption (A2):

$$\begin{aligned} \text{Cov}(X_i, u_i) &\equiv E\{[X_i - E(X_i)][u_i - E(u_i)]\} && \text{by definition} \\ &= E\{[X_i - E(X_i)]u_i\} && \text{since } E(u_i) = 0 \text{ by A2 and A2-1} \\ &= E[X_i u_i - E(X_i)u_i] \\ &= E(X_i u_i) - E(X_i)E(u_i) && \text{since } E(X_i) \text{ is a constant} \\ &= E(X_i u_i) && \text{since } E(u_i) = 0 \text{ by A2 and A2-1.} \end{aligned}$$

2. **Implication (A2-2)** states that the **population random error terms u_i** have *zero covariance with*, or are *uncorrelated with*, the **corresponding population regressor values X_i** . This assumption means that there exists *no linear association between u_i and X_i* .

Note that *zero covariance between X_i and u_i* implies *zero correlation between X_i and u_i* , since the *simple correlation coefficient between X_i and u_i* , denoted as $\rho(X_i, u_i)$, is defined as

$$\rho(X_i, u_i) \equiv \frac{\text{Cov}(X_i, u_i)}{\sqrt{\text{Var}(X_i)\text{Var}(u_i)}} = \frac{\text{Cov}(X_i, u_i)}{\text{sd}(X_i)\text{sd}(u_i)}.$$

From this definition of $\rho(X_i, u_i)$, it is obvious that $\text{Cov}(X_i, u_i) = 0$ implies that $\rho(X_i, u_i) = 0$, i.e.,

$$\text{Cov}(X_i, u_i) = 0 \Rightarrow \rho(X_i, u_i) = 0.$$

- **Implication 3 of A2.** Assumption A2 implies that the **conditional mean of the population Y_i values** corresponding to a given value X_i of the regressor X **equals** the **population regression function (PRF)**, $f(X_i) = \beta_0 + \beta_1 X_i$:

$$E(u_i | X_i) = 0 \Rightarrow E(Y_i | X_i) = f(X_i) = \beta_0 + \beta_1 X_i \quad (\text{A2-3})$$

$$E(u_i | X_i) = 0 \Rightarrow E(Y_i | X_i) = f(X_i) = \beta_0 + \beta_1 X_i \quad \forall i. \quad (\text{A2-3})$$

Proof: Take the conditional expectation of the PRE (A1) for some given X_i :

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad (i = 1, \dots, N) \quad (\text{A1})$$

$$\begin{aligned} E(Y_i | X_i) &= E(\beta_0 + \beta_1 X_i + u_i | X_i) \\ &= E(\beta_0 + \beta_1 X_i | X_i) \quad \text{since } E(u_i | X_i) = 0 \text{ by assumption A2} \\ &= \beta_0 + \beta_1 X_i \quad \text{since } E(\beta_0 + \beta_1 X_i | X_i) = \beta_0 + \beta_1 X_i. \end{aligned}$$

- **Meaning of the Zero Conditional Mean Error Assumption A2**

Each value X_i of X identifies a segment or subset of the relevant population. For each of these population segments or subsets, assumption A2 says that the mean of the random error u is zero. In other words, for each population segment, positive and negative values of u "cancel out" so that the average value of the random errors u_i for each population value X_i of X equals zero.

Assumption A2 rules out both **linear dependence and nonlinear dependence between X and u** ; that is, **it requires that X and u be statistically independent.**

- The absence of linear dependence between X and u means that **X and u are uncorrelated**, or equivalently that **X and u have zero covariance**.
- But linear independence between X and u is not sufficient to guarantee the satisfaction of assumption A2. It is possible for X and u to be both uncorrelated, or linearly independent, and nonlinearly related.
- Assumption A2 therefore also requires that there be **no nonlinear relationship between X and u** .

- **Violations of the Zero Conditional Mean Error Assumption A2**

- The random error term u represents all the **unknown, unobservable and unmeasured variables** other than the regressor X that determine the population values of the dependent variable Y .
- Anything that causes the random error u to be correlated with the regressor X will violate assumption A2:

$$\text{Cov}(X, u) \neq 0 \text{ or } \rho(X, u) \neq 0 \Rightarrow E(u|X) \neq 0.$$

If X and u are correlated, then $E(u|X)$ must depend on X and so cannot be zero.

Note that the **converse is not true**:

$$\text{Cov}(X, u) = 0 \text{ or } \rho(X, u) = 0 \text{ does not imply that } E(u|X) = 0.$$

Reason: $\text{Cov}(X, u)$ measures only **linear dependence** between u and X . But any **nonlinear dependence** between u and X will also cause $E(u|X)$ to depend on X , and hence to differ from zero. So $\text{Cov}(X, u) = 0$ is not enough to insure that assumption A2 is satisfied.

- **Common causes of correlation or dependence between X and u** -- i.e., common causes of violations of assumption A2.
 1. **Incorrect specification of the functional form of the relationship between Y and X .**

Examples: Using Y as the dependent variable when the true model has $\ln(Y)$ as the dependent variable. Or using X as the independent variable when the true model has $\ln(X)$ as the independent variable.
 2. **Omission of relevant variables that are correlated with X .**
 3. **Measurement errors in X .**
 4. **Joint determination of X and Y .**

Assumption A3: The Assumption of *Constant Error Variances*
The Assumption of *Homoskedastic Errors*
The Assumption of *Homoskedasticity*

The *conditional* variances of the random error terms u_i are identical for all observations (i.e., for all population values X_i of X), and equal the same finite positive constant σ^2 for all i :

$$\text{Var}(u | X) = E(u^2 | X) = \sigma^2 > 0 \quad (\text{A3})$$

or

$$\text{Var}(u_i | X_i) = E(u_i^2 | X_i) = \sigma^2 > 0 \quad \forall i \text{ (for all } X_i \text{ values)} \quad (\text{A3})$$

where σ^2 is a *finite positive (unknown) constant*.

- The **first equality in (A3)** follows from the definition of the conditional variance of u_i and assumption (A2):

$$\begin{aligned} \text{Var}(u_i | X_i) &\equiv E\{[u_i - E(u_i | X_i)]^2 | X_i\} \quad \text{by definition} \\ &= E\{[u_i - 0]^2 | X_i\} \quad \text{because } E(u_i | X_i) = 0 \text{ by assumption A2} \\ &= E(u_i^2 | X_i). \end{aligned}$$

- Implication 1 of A3:** Assumption A3 implies that the *unconditional variance of the random error u* is also equal to σ^2 :

$$\text{Var}(u_i) = E[(u_i - E(u_i))^2] = E(u_i^2) = \sigma^2.$$

where $\text{Var}(u_i) = E(u_i^2)$ because $E(u_i) = 0$ by A2-1.

By assumptions A2 and A3, $E(u_i^2 | X) = \sigma^2$.

By the law of iterated expectations, $E[E(u_i^2 | X_i)] = E(u_i^2)$.

Thus,

$$\text{Var}(u_i) = E(u_i^2) = E[E(u_i^2 | X_i)] = E[\sigma^2] = \sigma^2.$$

-
- **Implication 2 of A3:** Assumption A3 implies that the conditional variance of the population values Y_i of the regressand Y corresponding to any given value X_i of the regressor X equals the constant conditional error variance σ^2 :

$$\text{Var}(u_i | X_i) = \sigma^2 \quad \forall i \quad \Rightarrow \quad \text{Var}(Y_i | X_i) = \sigma^2 \quad \forall i \quad (\text{A3-2})$$

Proof: Start with the definition of the conditional variance of Y_i for some given X_i :

$$\begin{aligned} \text{Var}(Y_i | X_i) &\equiv E\{[Y_i - E(Y_i | X_i)]^2 | X_i\} && \text{by definition} \\ &= E\{[Y_i - \beta_0 - \beta_1 X_i]^2 | X_i\} && \text{since } E(Y_i | X_i) = \beta_0 + \beta_1 X_i \text{ by A2} \\ &= E(u_i^2 | X_i) && \text{since } u_i = Y_i - \beta_0 - \beta_1 X_i \text{ by A1} \\ &= \sigma^2 && \text{since } E(u_i^2 | X_i) = \sigma^2 \text{ by assumption A3.} \end{aligned}$$

- **Meaning of the Homoskedasticity Assumption A3**

- For each population value X_i of X , there is a **conditional distribution of random errors**, and a corresponding **conditional distribution of population Y values**.
- Assumption A3 says that the **variance of the random errors for $X = X_i$** is the **same** as the **variance of the random errors for any other regressor value $X = X_s$** (for all $X_s \neq X_i$). That is, the **variances of the conditional random error distributions** corresponding to each population value of X are all **equal to the same finite positive constant σ^2** .

$$\text{Var}(u_i | X_i) = \text{Var}(u_s | X_s) = \sigma^2 > 0 \quad \text{for all } X_s \neq X_i.$$

- Implication A3-2 says that the **variance of the population Y values for $X = X_i$** is the **same** as the **variance of the population Y values for any other regressor value $X = X_s$** (for all $X_s \neq X_i$). The **conditional distributions of the population Y values** around the PRF have the **same constant variance σ^2** for all population values of X .

$$\text{Var}(Y_i | X_i) = \text{Var}(Y_s | X_s) = \sigma^2 > 0 \quad \text{for all } X_s \neq X_i.$$

Assumption A4: **The Assumption of Zero Error Covariances**
 The Assumption of Nonautoregressive Errors
 The Assumption of Nonautocorrelated Errors

Consider the random error terms u_i and u_s ($i \neq s$) corresponding to two different population values X_i and X_s of the regressor X , where $X_i \neq X_s$. This assumption states that u_i and u_s have zero conditional covariance:

$$\text{Cov}(u_i, u_s | X_i, X_s) = E(u_i u_s | X_i, X_s) = 0 \quad \forall i \neq s \text{ (for all } X_i \neq X_s) \quad (\text{A4})$$

- The **first equality in (A4)** follows from the definition of the conditional covariance of u_i and u_s and assumption (A2):

$$\begin{aligned} \text{Cov}(u_i, u_s | X_i, X_s) &\equiv E\{[u_i - E(u_i | X_i)][u_s - E(u_s | X_s)] | X_i, X_s\} \quad \text{by definition} \\ &= E(u_i u_s | X_i, X_s) \quad \text{since } E(u_i | X_i) = E(u_s | X_s) = 0 \text{ by A2.} \end{aligned}$$

- The **second equality in (A4)** states the assumption that all pairs of error terms corresponding to different values of X have zero covariance.
- **Implication of A4:** Assumption A4 implies that the conditional covariance between the population values Y_i of Y when $X = X_i$ and the population values Y_s of Y when $X = X_s$ where $X_i \neq X_s$ is equal to zero:

$$\text{Cov}(u_i, u_s | X_i, X_s) = 0 \Rightarrow \text{Cov}(Y_i, Y_s | X_i, X_s) = 0 \quad \forall X_i \neq X_s.$$

Proof:

- (1) Begin with the definition of the conditional covariance for Y_i and Y_s for given X_i and X_s values:

$$\begin{aligned}\text{Cov}(Y_i, Y_s | X_i, X_s) &\equiv E\{[Y_i - E(Y_i | X_i)][Y_s - E(Y_s | X_s)] | X_i, X_s\} \\ &= E(u_i u_s | X_i, X_s)\end{aligned}$$

since

$$Y_i - E(Y_i | X_i) = Y_i - \beta_0 - \beta_1 X_i = u_i \quad \text{by assumptions A1 and A2,}$$

and similarly

$$Y_s - E(Y_s | X_s) = Y_s - \beta_0 - \beta_1 X_s = u_s \quad \text{by assumptions A1 and A2.}$$

- (2) Therefore

$$\text{Cov}(Y_i, Y_s | X_i, X_s) = E(u_i u_s | X_i, X_s) = 0 \quad \text{by assumption A4.}$$

- **Meaning of A4:** Assumption A4 states that the random error terms u_i corresponding to $X = X_i$ have **zero covariance with**, or are **uncorrelated with**, the random error terms u_s corresponding to any other regressor value $X = X_s$, where $X_i \neq X_s$. Equivalently, assumption A4 states that the **population values Y_i of Y** corresponding to $X = X_i$ have **zero covariance with**, or are **uncorrelated with**, the **population values Y_s of Y** corresponding to $X = X_s$ for any distinct pair of regressor values $X_i \neq X_s$. This means there is **no systematic linear dependence or association between u_i and u_s , or between Y_i and Y_s** , where i and s correspond to different observations (that is, to different regressor values $X_i \neq X_s$).
- The **assumption of zero covariance, or zero correlation**, between each pair of distinct observations is **weaker** than the **assumption of independent random sampling A5** from an underlying population.
- The assumption of independent random sampling implies that the sample observations are statistically independent. The assumption of statistically independent observations is **sufficient** for the assumption of zero covariance between observations, but is stronger than necessary.

4. Properties of the Sample Data

Assumption A5: Random Sampling or Independent Random Sampling

The **sample data** consist of N *randomly selected observations* on the regressand Y and the regressor X , the two observable variables in the PRE described by A1.

In other words, the sample observations are **randomly selected** from the underlying population; they are a **random subset** of the **population data points**.

These N randomly selected sample observations on Y and X are written as the N pairs

$$\begin{aligned}\text{Sample data} &\equiv [(Y_1, X_1), (Y_2, X_2), \dots (Y_N, X_N)] \\ &\equiv (Y_i, X_i) \quad i = 1, \dots, N.\end{aligned}$$

- Implications of the Random Sampling Assumption A5

The **assumption of random sampling** implies that **the sample observations are statistically independent**.

1. It thus means that the error terms u_i and u_s are **statistically independent**, and hence have zero covariance, for any two observations i and s .

$$\text{Random sampling} \Rightarrow \text{Cov}(u_i, u_s | X_i, X_s) = \text{Cov}(u_i, u_s) = 0 \quad \forall i \neq s.$$

2. It also means that the dependent variable values Y_i and Y_s are **statistically independent**, and hence have zero covariance, for any two observations i and s .

$$\text{Random sampling} \Rightarrow \text{Cov}(Y_i, Y_s | X_i, X_s) = \text{Cov}(Y_i, Y_s) = 0 \quad \forall i \neq s.$$

The assumption of random sampling is therefore sufficient for assumption A4 of zero covariance between observations, but is stronger than is necessary for A4.

- **When is the Random Sampling Assumption A5 Appropriate?**

The random sampling assumption is usually appropriate for ***cross-sectional regression models***, i.e., for regression models formulated for ***cross section data***.

- ***Definition:*** A **cross-sectional data set** consists of a sample of observations on individual economic agents or other units taken at a **single point in time** or over a **single period of time**.
- A **distinguishing characteristic** of any cross-sectional data set is that the individual observations have **no natural ordering**.
- A **common, almost universal characteristic** of cross-sectional data sets is that they usually are constructed by **random sampling** from underlying populations.

The random sampling assumption is hardly ever appropriate for ***time-series regression models***, i.e., for regression models formulated for ***time series data***.

- ***Definition:*** A **time-series data set** consists of a sample of observations on one or more variables over several successive periods or intervals of time.
- A **distinguishing characteristic** of any time-series data set is that the observations have a natural ordering -- specifically a **chronological ordering**.
- A **common, almost universal characteristic** of time-series data sets is that the **sample observations exhibit a high degree of time dependence**, and therefore cannot be assumed to be generated by random sampling.

Assumption A6: The number of sample observations N is greater than the number of unknown parameters K :

number of sample observations $>$ number of unknown parameters

$$N > K. \quad (A6)$$

- **Meaning of A6:** Unless this assumption is satisfied, it is not possible to compute from a given sample of N observations estimates of all the unknown parameters in the model.

Assumption A7: Nonconstant Regressor

The sample values X_i of the regressor X in the sample (and hence in the population) are not all the same; i.e., they are not constant:

$$X_i \neq c \quad \forall i = 1, \dots, N \quad \text{where } c \text{ is a constant.} \quad (A7)$$

- **Technical Form of A7:** Assumption A7 requires that the sample variance of the regressor values X_i ($i = 1, \dots, N$) must be a *finite positive number* for any sample size N ; i.e.,

$$\text{sample variance of } X_i \equiv \text{Var}(X_i) = \frac{\sum_i (X_i - \bar{X})^2}{N-1} = s_x^2 > 0,$$

where $s_x^2 > 0$ is a *finite positive number*.

- **Meaning of A7:** Assumption A7 requires that the regressor sample values X_i take at least two different values in any given sample.

Unless this assumption is satisfied, it is not possible to compute from the sample data an estimate of the effect on the regressand Y of changes in the value of the regressor X . In other words, to calculate the effect of changes in X on Y , the sample values X_i of the regressor X must vary across observations in any given sample.

Assumption A8: No Perfect Multicollinearity

The sample values of the regressors in a multiple regression model do not exhibit perfect multicollinearity.

This assumption is relevant only in multiple regression models that contain two or more non-constant regressors.

Its nature is examined later in the context of multiple regression models.

